$$\mathcal{L}_{\mathrm{bc}} = \|a_t^S - a_t^T\|_2^2$$

$$\mathcal{L}_{\mathrm{re}} = \|[\widehat{o}_t^e, \widehat{s}_t^p] - [o_t^e, s_t^p]\|_2^2$$

$$\mathcal{L} = \mathcal{L}_{\mathrm{bc}} + 0.5\mathcal{L}_{\mathrm{re}}$$